

A Self-Contained Norm-Oblivious Proof for Correlation Clustering

Complete resolution of the formalized simultaneous all- ℓ_p statement

June 24, 2026

Status of the result. The argument below is a complete proof of the formal mathematical statement: it produces one clustering that is a constant-factor approximation simultaneously for every ℓ_p objective. Thus, relative to that formal statement, this is a *solution*, not merely partial progress.

The result is not new in the literature. Davies–Moseley–Newman had already proved the existence of a simultaneous constant-factor approximation for all ℓ_p norms [2], and Cao–Li–Ye subsequently proved the stronger simultaneous 63.3-approximation theorem for every monotone symmetric norm [1]. The construction and charging argument below are a self-contained presentation of the latter approach. The exact expression $35 + 20\sqrt{2}$ is the optimized form of the displayed bound, whose decimal value is approximately 63.2843.

Abstract

For an unweighted complete signed graph and a clustering $\mathcal{C}$, let $y_{\mathcal{C}}(u)$ be the number of disagreements incident to vertex u . We construct one clustering $\hat{\mathcal{C}}$, independent of the objective norm, such that

$$\|y_{\hat{\mathcal{C}}}\|_p \leq (35 + 20\sqrt{2}) \min_{\mathcal{C}} \|y_{\mathcal{C}}\|_p \quad \text{for every } p \in [1, \infty].$$

In fact, the same clustering satisfies the analogous guarantee for every monotone symmetric norm. The proof has three ingredients: a positive-neighborhood-overlap metric, a simultaneous top- k charging bound for its fractional disagreement vector, and a ball-growing rounding procedure whose loss is bounded separately at every vertex. Every step is proved explicitly, including the triangle inequality, the coefficient budgets in the charging argument, and the coordinatewise factor-5 rounding guarantee.

1 Problem statement and theorem

Let V be a set of n vertices. Every unordered pair $uv \in \binom{V}{2}$ is labeled either positive or negative; write E^+ and E^- for the two classes. A clustering $\mathcal{C}$ is a partition of V . A positive pair is a disagreement when its endpoints lie in different clusters, while a negative pair is a disagreement when its endpoints lie in the same cluster. Define the disagreement vector $y_{\mathcal{C}} \in \mathbb{R}_+^V$ by

$$y_{\mathcal{C}}(u) := \#\{v \in V \setminus \{u\} : uv \text{ is a disagreement under } \mathcal{C}\}.$$

For $p \in [1, \infty]$, let

$$\text{OPT}_p := \min_{\mathcal{C}} \|y_{\mathcal{C}}\|_p.$$

More generally, for a norm Φ on $\mathbb{R}^V$, set

$$\text{OPT}_\Phi := \min_C \Phi(y_C).$$

A norm is called symmetric if it is invariant under coordinate permutations and sign changes. Such norms include every ℓ_p norm and every top- k norm.

Theorem 1.1 (Norm-oblivious clustering). *For every complete signed graph there is a clustering $\hat{C}$, computable in polynomial time, such that*

$$\Phi(y_{\hat{C}}) \leq (35 + 20\sqrt{2}) \text{OPT}_\Phi$$

for every symmetric norm Φ on $\mathbb{R}^V$. In particular,

$$\|y_{\hat{C}}\|_p \leq (35 + 20\sqrt{2}) \text{OPT}_p \quad \text{for every } p \in [1, \infty].$$

The clustering $\hat{C}$ is independent of Φ and of p .

The proof is organized around the top- k functionals. For $z \in \mathbb{R}_+^V$, let $z_1^\downarrow \geq \dots \geq z_n^\downarrow$ be the coordinates of z in nonincreasing order, and define

$$\text{Top}_k(z) := \sum_{i=1}^k z_i^\downarrow = \max_{U \subseteq V: |U|=k} \sum_{u \in U} z(u).$$

Also write

$$\text{OPT}_k^{\text{top}} := \min_C \text{Top}_k(y_C).$$

We shall construct one feasible metric x whose fractional disagreement vector f_x satisfies

$$\text{Top}_k(f_x) \leq (7 + 4\sqrt{2}) \text{OPT}_k^{\text{top}} \quad (k = 1, \dots, n),$$

and then round it coordinatewise with factor 5.

2 Three general tools

2.1 A coefficient-budget lemma

Lemma 2.1 (Coefficient budget). *Let $c, q \in \mathbb{R}_+^V$, let $k \in \{1, \dots, n\}$, and suppose*

$$\|q\|_\infty \leq \alpha, \quad \|q\|_1 \leq \alpha k.$$

Then

$$q \cdot c \leq \alpha \text{Top}_k(c).$$

Proof. Sort the coordinates of c in nonincreasing order. Under the constraints $0 \leq q_i \leq \alpha$ and $\sum_i q_i \leq \alpha k$, the scalar product $q \cdot c$ is maximized by assigning weight α to the k largest coordinates of c and zero to the remaining coordinates. Hence

$$q \cdot c \leq \alpha \sum_{i=1}^k c_i^\downarrow = \alpha \text{Top}_k(c).$$

□

2.2 Top- k domination implies domination for every symmetric norm

Lemma 2.2 (Ky Fan domination, in the form needed here). *Let $a, b \in \mathbb{R}_+^n$ and let $\alpha \geq 0$. If*

$$\text{Top}_k(a) \leq \alpha \text{Top}_k(b) \quad (k = 1, \dots, n),$$

then

$$\Phi(a) \leq \alpha \Phi(b)$$

for every symmetric norm Φ on $\mathbb{R}^n$.

Proof. Let Φ^* be the dual norm. The dual of a symmetric norm is symmetric. Fix any $z \in \mathbb{R}^n$ with $\Phi^*(z) \leq 1$. Since $a \geq 0$,

$$z \cdot a \leq |z| \cdot a \leq |z|^\downarrow \cdot a^\downarrow,$$

where the second inequality is the rearrangement inequality. Symmetry gives $\Phi^*(|z|^\downarrow) = \Phi^*(z) \leq 1$. Put $w_i = |z|_i^\downarrow$ and $w_{n+1} = 0$. Abel summation yields

$$\begin{aligned} w \cdot a^\downarrow &= \sum_{k=1}^n (w_k - w_{k+1}) \text{Top}_k(a) \\ &\leq \alpha \sum_{k=1}^n (w_k - w_{k+1}) \text{Top}_k(b) \\ &= \alpha w \cdot b^\downarrow \leq \alpha \Phi(b), \end{aligned}$$

where the final inequality is duality. Taking the supremum over the dual unit ball proves the claim. $\square$

2.3 A coordinatewise factor-5 rounding lemma

A symmetric array $x = (x_{uv})_{u,v \in V}$ is a feasible correlation-clustering metric if

$$x_{uu} = 0, \quad 0 \leq x_{uv} \leq 1, \quad x_{uv} \leq x_{uw} + x_{wv} \quad (u, v, w \in V).$$

Its fractional disagreement at u is

$$f_x(u) := \sum_{v:uv \in E^+} x_{uv} + \sum_{v:uv \in E^-} (1 - x_{uv}). \quad (1)$$

The following is the local ball-growing rounding lemma used by Kalhan–Makarychev–Zhou [3]. We include a proof because its coordinatewise conclusion is what makes a norm-oblivious argument possible.

Lemma 2.3 (Local ball rounding). *Given any feasible metric x , one can construct a clustering $\mathcal{C}_x$ such that*

$$y_{\mathcal{C}_x}(u) \leq 5f_x(u) \quad \text{for every } u \in V.$$

Proof. Set $r = 1/5$. During the algorithm let S be the set of active vertices and define

$$B_S(w, \rho) := \{u \in S : x_{wu} \leq \rho\}, \quad L_S(w) := \sum_{u \in B_S(w, r)} (r - x_{wu}).$$

Choose a vertex $w \in S$ maximizing $L_S(w)$, output $C = B_S(w, 2r)$ as the next cluster, remove C from S , and repeat.

We prove the guarantee round by round. For an unordered pair uv , define its fractional loss

$$\ell_x(u, v) = \begin{cases} x_{uv}, & uv \in E^+, \\ 1 - x_{uv}, & uv \in E^-. \end{cases}$$

A pair is removed in the current round if at least one endpoint lies in C . For a removed pair, its final agreement/disagreement status is already determined. Let $A(u, v)$ be the indicator that uv is a disagreement in the final clustering, and define

$$P(u, v) = \begin{cases} \ell_x(u, v) - rA(u, v), & uv \text{ is removed in this round,} \\ 0, & \text{otherwise.} \end{cases}$$

We claim that, in every round and for every active $u \in S$,

$$\sum_{v \in S \setminus \{u\}} P(u, v) \geq 0. \quad (2)$$

First consider a negative pair. Such a pair can be a disagreement only if both endpoints enter C . Then $x_{uv} \leq x_{uw} + x_{vw} \leq 4r$, so $1 - x_{uv} \geq 1 - 4r = r$. Thus $P(u, v) \geq 0$ for every negative pair.

It remains to consider positive pairs. If $x_{uw} \leq r$, then $u \in C$; whenever a positive pair uv is cut, $v \notin C$, and hence

$$x_{uv} \geq x_{vw} - x_{uw} \geq r.$$

Similarly, if $x_{uw} \geq 3r$, then a cut positive pair has $v \in C$, so $x_{uv} \geq x_{uw} - x_{vw} \geq r$. In both regimes every positive-pair contribution is nonnegative.

Suppose now that $r < x_{uw} < 3r$. Split the sum according to whether $v \in B_S(w, r)$. For every such v one has

$$P(u, v) \geq r - x_{vw}. \quad (3)$$

Indeed, if $x_{uv} \leq 2r$, then $u, v \in C$. For a positive pair, $P(u, v) = x_{uv} \geq x_{uw} - x_{vw} \geq r - x_{vw}$; for a negative pair,

$$P(u, v) = 1 - x_{uv} - r \geq 1 - (2r + x_{vw}) - r = 2r - x_{vw} \geq r - x_{vw}.$$

If $x_{uv} > 2r$, then $u \notin C$ and $v \in C$. For a positive pair,

$$P(u, v) = x_{uv} - r \geq x_{uv} - x_{vw} - r \geq r - x_{vw},$$

whereas for a negative pair,

$$P(u, v) = 1 - x_{uv} \geq 1 - (3r + x_{vw}) = 2r - x_{vw} \geq r - x_{vw}.$$

Consequently, the total contribution from $B_S(w, r)$ is at least $L_S(w)$.

For $v \notin B_S(w, r)$, negative pairs contribute nonnegatively, while a positive pair contributes one of 0, x_{uv} , and $x_{uv} - r$. Hence

$$P(u, v) \geq \min\{x_{uv} - r, 0\}.$$

Adding the omitted nonpositive terms, including the harmless virtual term $v = u$, gives

$$\sum_{v \notin B_S(w, r)} P(u, v) \geq \sum_{v \in S} \min\{x_{uv} - r, 0\} = -L_S(u).$$

Since w maximizes L_S , inequality (2) follows.

Every unordered pair is removed in exactly one round. Fixing u and summing (2) over all rounds in which u remains active gives

$$f_x(u) - ry_{C_x}(u) \geq 0.$$

Since $r = 1/5$, this is the desired coordinatewise bound. $\square$

3 The norm-oblivious metric

For neighborhood bookkeeping, give every vertex a positive self-loop. Thus

$$N(u) := \{u\} \cup \{v \neq u : uv \in E^+\}, \quad d(u) := |N(u)|, \quad M_{uv} := \max\{d(u), d(v)\}.$$

Fix a parameter $\beta \in (0, 1)$. Retain a positive nonloop pair uv if

$$|N(u) \triangle N(v)| \leq \beta M_{uv}, \tag{4}$$

and retain every self-loop. Let H be the resulting graph and let $N_H(u)$ be the retained neighborhood of u . Define

$$x_{uu} = 0, \quad x_{uv} = 1 - \frac{|N_H(u) \cap N_H(v)|}{M_{uv}} \quad (u \neq v). \tag{5}$$

The construction depends only on the signed instance and β .

We shall repeatedly use the following consequence of retention. If uv is retained, then

$$\min\{d(u), d(v)\} \geq (1 - \beta) \max\{d(u), d(v)\}. \tag{6}$$

Indeed, $|d(u) - d(v)| \leq |N(u) \triangle N(v)| \leq \beta M_{uv}$.

Proposition 3.1. *The array x defined in (5) is a feasible metric with values in $[0, 1]$.*

Proof. Symmetry and $x_{uu} = 0$ are immediate. Moreover,

$$|N_H(u) \cap N_H(v)| \leq \min\{d(u), d(v)\} \leq M_{uv},$$

so $0 \leq x_{uv} \leq 1$.

It remains to prove the triangle inequality. Take distinct u, v, w and assume, without loss of generality, that $d(v) \geq d(w)$; hence $M_{vw} = d(v)$. Write

$$A = N_H(u), \quad B = N_H(v), \quad D = N_H(w).$$

Because $M_{uv} \geq d(v)$ and $M_{uv}, M_{uw} \geq d(u)$,

$$\begin{aligned} x_{uv} + x_{uw} - x_{vw} &= 1 + \frac{|B \cap D|}{d(v)} - \frac{|A \cap B|}{M_{uv}} - \frac{|A \cap D|}{M_{uw}} \\ &\geq 1 + \frac{|B \cap D| - |A \cap B|}{M_{uv}} - \frac{|A \cap D|}{M_{uw}} \\ &\geq 1 - \frac{|A \setminus D|}{M_{uv}} - \frac{|A \cap D|}{M_{uw}} \\ &\geq 1 - \frac{|A \setminus D| + |A \cap D|}{d(u)} \\ &= 1 - \frac{|N_H(u)|}{d(u)} \geq 0. \end{aligned}$$

The second inequality uses $|A \cap B| - |B \cap D| \leq |A \setminus D|$. This proves $x_{vw} \leq x_{uv} + x_{uw}$. $\square$

4 A simultaneous bound for every top- k objective

Fix $k \in \{1, \dots, n\}$ and an arbitrary comparison clustering $\mathcal{C}$. Put

$$c(u) := y_{\mathcal{C}}(u).$$

Choose $U \subseteq V$, $|U| = k$, such that

$$\text{Top}_k(f_x) = \sum_{u \in U} f_x(u).$$

All sums below are incidence sums: an unordered pair contributes once for each endpoint lying in U .

Partition the nonloop pairs into the following five classes:

$$\begin{aligned} D^+ &= \{uv \in E^+ : \mathcal{C}(u) \neq \mathcal{C}(v)\}, \\ D^- &= \{uv \in E^- : \mathcal{C}(u) = \mathcal{C}(v)\}, \\ B &= \{uv \in E^+ \setminus E(H) : \mathcal{C}(u) = \mathcal{C}(v)\}, \\ A &= \{uv \in E(H) \cap E^+ : \mathcal{C}(u) = \mathcal{C}(v)\}, \\ S &= \{uv \in E^- : \mathcal{C}(u) \neq \mathcal{C}(v)\}. \end{aligned}$$

Thus $D^+ \cup D^-$ is the set of disagreements of $\mathcal{C}$; A and B are the uncut positive pairs, split according to whether they were retained; and S is the set of cut negative pairs. For $F \in \{A, B\}$, let $F(u)$ be the set of F -neighbors of u , and write

$$a_u := |A(u)|, \quad b_u := |B(u)|.$$

Since $A(u)$ and $B(u)$ are disjoint subsets of $N(u)$,

$$a_u + b_u \leq d(u). \tag{7}$$

4.1 The direct disagreement classes

Since every fractional pair loss lies in $[0, 1]$,

$$\sum_{\substack{u \in U \\ uv \in D^+}} x_{uv} + \sum_{\substack{u \in U \\ uv \in D^-}} (1 - x_{uv}) \leq \sum_{u \in U} c(u) \leq \text{Top}_k(c). \tag{8}$$

4.2 Uncut positive pairs

We first establish two edgewise estimates.

Lemma 4.1 (Removed uncut positive edge). *If $uv \in B$, then*

$$x_{uv} \leq \frac{c(u) + c(v)}{\beta M_{uv}}. \tag{9}$$

Proof. The endpoints u, v lie in the same cluster. For each $w \in N(u) \Delta N(v)$, exactly one of uw and vw is positive. If w lies in the cluster of u and v , then the negative one is a disagreement; if w lies outside that cluster, then the positive one is a disagreement. Distinct vertices w yield distinct incident disagreement edges. Therefore

$$c(u) + c(v) \geq |N(u) \Delta N(v)| > \beta M_{uv},$$

where the strict inequality holds because uv was not retained. Since $x_{uv} \leq 1$, inequality (9) follows. $\square$

Lemma 4.2 (Retained uncut positive edge). *If $uv \in A$, then*

$$x_{uv} \leq \frac{c(u) + c(v)}{M_{uv}} + \sum_{w \in B(u)} \frac{c(u) + c(w)}{\beta M_{uv} M_{uw}} + \sum_{w \in B(v)} \frac{c(v) + c(w)}{\beta M_{uv} M_{vw}}. \quad (10)$$

Proof. Let $z \in \{u, v\}$ have $d(z) = M_{uv}$, and let z' be the other endpoint. Since uv is retained,

$$x_{uv} = \frac{|N(z) \setminus (N_H(u) \cap N_H(v))|}{M_{uv}}.$$

Consider a vertex w in the numerator. The edge zw is positive. If zw is not retained, then either $zw \in B$ or zw is a positive disagreement incident to z . If zw is retained but $z'w$ is not, then either zw is a positive disagreement incident to z , or w lies in the common cluster of u, v ; in the latter case, $z'w$ is either an edge of B or a negative disagreement incident to z' . Charging each such w to the corresponding incident disagreement or removed edge gives

$$x_{uv} \leq \frac{c(u) + c(v) + b_u + b_v}{M_{uv}}. \quad (11)$$

If $uv \in B$, the proof of Lemma 4.1 gives the stronger inequality $c(u) + c(v) > \beta M_{uv}$, and hence

$$1 \leq \frac{c(u) + c(v)}{\beta M_{uv}}.$$

Use this inequality for every edge counted by b_u and b_v in (11). The result is (10). $\square$

Proposition 4.3 (Charge from uncut positive pairs). *The contribution of all pairs in $A \cup B$ to $\sum_{u \in U} f_x(u)$ is at most*

$$\frac{4}{\beta} \text{Top}_k(c).$$

Proof. For $F \in \{A, B\}$, write

$$\mathcal{I}_F := \{(u, v) : u \in U, uv \in F\}$$

for the oriented F -incidences whose first endpoint belongs to U .

The class B . Sum (9) over $\mathcal{I}_B$. The result is at most $q^B \cdot c$ for a nonnegative coefficient vector q^B . A coordinate $c(r)$ can occur with r as the first or the second endpoint, and every relevant denominator is at least $d(r)$. Therefore

$$q^B(r) \leq \frac{2b_r}{\beta d(r)}. \quad (12)$$

Each oriented edge $(u, v) \in \mathcal{I}_B$ contributes total coefficient $2/(\beta M_{uv}) \leq 2/(\beta d(u))$, so

$$\|q^B\|_1 \leq \frac{2}{\beta} \sum_{u \in U} \frac{b_u}{d(u)}. \quad (13)$$

The class A. Sum (10) over $\mathcal{I}_A$, and split the resulting coefficient vector as $q^A = q^0 + q^1 + q^2$, according to the three terms on the right side of (10).

For the first term, a coordinate can appear as either endpoint of an A -edge, giving

$$q^0(r) \leq \frac{2a_r}{d(r)}, \quad \|q^0\|_1 \leq 2 \sum_{u \in U} \frac{a_u}{d(u)}. \quad (14)$$

Consider next the term involving $B(u)$. The coefficient of $c(u)$ for a fixed first endpoint u is at most $a_u b_u / (\beta d(u)^2)$. If a coordinate $c(r)$ appears in the role of $c(w)$, then

$$\begin{aligned} & \sum_{u \in U: ur \in B} \sum_{v \in A(u)} \frac{1}{\beta M_{uv} M_{ur}} \\ & \leq \sum_{u \in U: ur \in B} \frac{1}{\beta M_{ur}} \left(\sum_{v \in A(u)} \frac{1}{M_{uv}} \right) \leq \sum_{u \in U: ur \in B} \frac{1}{\beta M_{ur}} \leq \frac{b_r}{\beta d(r)}. \end{aligned}$$

Thus

$$q^1(r) \leq \frac{a_r b_r}{\beta d(r)^2} + \frac{b_r}{\beta d(r)}. \quad (15)$$

Since each triple (u, v, w) contributes total coefficient $2/(\beta M_{uv} M_{uw})$, we also have

$$\|q^1\|_1 \leq \frac{2}{\beta} \sum_{u \in U} \frac{a_u b_u}{d(u)^2}. \quad (16)$$

For the term involving $B(v)$, the same pointwise argument gives

$$q^2(r) \leq \frac{a_r b_r}{\beta d(r)^2} + \frac{b_r}{\beta d(r)}. \quad (17)$$

For its ℓ_1 mass, use

$$\sum_{w \in B(v)} \frac{1}{M_{vw}} \leq \frac{b_v}{d(v)} \leq 1.$$

It follows that

$$\|q^2\|_1 \leq \frac{2}{\beta} \sum_{u \in U} \frac{a_u}{d(u)}. \quad (18)$$

Combining (14), (15), and (17),

$$q^A(r) \leq 2 \left(\frac{a_r}{d(r)} + \frac{a_r b_r}{\beta d(r)^2} + \frac{b_r}{\beta d(r)} \right). \quad (19)$$

Combining (14), (16), and (18),

$$\|q^A\|_1 \leq 2 \sum_{u \in U} \left(\frac{a_u}{d(u)} + \frac{a_u b_u}{\beta d(u)^2} + \frac{a_u}{\beta d(u)} \right). \quad (20)$$

The combined coefficient budget. Let $q = q^A + q^B$. For a fixed vertex r , put

$$a = \frac{a_r}{d(r)}, \quad b = \frac{b_r}{d(r)}.$$

Then $a, b \geq 0$ and $a + b \leq 1$ by (7). Equations (12) and (19) give

$$q(r) \leq \frac{2}{\beta}(\beta a + ab + 2b).$$

Since $a \leq 1 - b$,

$$\begin{aligned} \beta a + ab + 2b &\leq \beta(1 - b) + (1 - b)b + 2b \\ &= \beta + (3 - \beta)b - b^2 \\ &= 2 - (1 - b)(2 - \beta - b) \leq 2. \end{aligned}$$

Hence

$$\|q\|_\infty \leq \frac{4}{\beta}. \tag{21}$$

Similarly, (13) and (20) yield

$$\begin{aligned} \|q\|_1 &\leq \frac{2}{\beta} \sum_{u \in U} \left(\frac{b_u}{d(u)} + \beta \frac{a_u}{d(u)} + \frac{a_u b_u}{d(u)^2} + \frac{a_u}{d(u)} \right) \\ &\leq \frac{4k}{\beta}. \end{aligned}$$

For the last inequality, writing again $a = a_u/d(u)$ and $b = b_u/d(u)$, one has $b + \beta a + ab + a \leq 2(a + b) \leq 2$. Lemma 2.1, with $\alpha = 4/\beta$, proves the proposition. $\square$

4.3 Cut negative pairs

Proposition 4.4 (Charge from cut negative pairs). *The contribution of all pairs in S to $\sum_{u \in U} f_x(u)$ is at most*

$$\frac{2}{1 - \beta} \text{Top}_k(c).$$

Proof. For $uv \in S$,

$$1 - x_{uv} = \frac{|N_H(u) \cap N_H(v)|}{M_{uv}}.$$

Fix an oriented incidence (u, v) with $u \in U$ and expand the intersection over common retained neighbors w .

First suppose that w lies in the same cluster as u . Since u and v lie in different clusters, vw is a retained positive disagreement incident to w . By degree comparability, $d(v) \geq (1 - \beta)d(w)$, and therefore

$$M_{uv} \geq \max\{d(u), (1 - \beta)d(w)\}.$$

For fixed u, w , the possible vertices v correspond to distinct positive disagreements incident to w , so there are at most $c(w)$ of them. The total contribution of this case is at most

$$\sum_{u \in U} \sum_{\substack{w \in N_H(u) \\ \mathcal{C}(w) = \mathcal{C}(u)}} \frac{c(w)}{\max\{d(u), (1 - \beta)d(w)\}}. \tag{22}$$

Now suppose that w lies outside the cluster of u . Then uw is itself a retained positive disagreement incident to u . For fixed u, w , at most $|N_H(w)| \leq d(w)$ vertices v can occur, and every denominator is at least $d(u)$. Since uw is retained, (6) gives $d(w)/d(u) \leq 1/(1 - \beta)$. There are at most $c(u)$ possible vertices w , so this case contributes at most

$$\sum_{u \in U} \frac{c(u)}{1 - \beta}. \quad (23)$$

The right sides of (22) and (23) equal $q^S \cdot c$ for a nonnegative coefficient vector q^S . For a fixed coordinate r , the first part contributes at most

$$\frac{|N_H(r)|}{(1 - \beta)d(r)} \leq \frac{1}{1 - \beta},$$

and the second part contributes at most $1/(1 - \beta)$. Hence

$$\|q^S\|_\infty \leq \frac{2}{1 - \beta}. \quad (24)$$

For the ℓ_1 mass, sum (22) by its first endpoint $u \in U$ and use the denominator lower bound $d(u)$; this gives at most $\sum_{u \in U} |N_H(u)|/d(u) \leq k$. The second part has mass $k/(1 - \beta)$. Consequently,

$$\|q^S\|_1 \leq k + \frac{k}{1 - \beta} \leq \frac{2k}{1 - \beta}. \quad (25)$$

Lemma 2.1, with $\alpha = 2/(1 - \beta)$, proves the proposition. $\square$

4.4 The simultaneous fractional guarantee

Theorem 4.5. *For every $k \in \{1, \dots, n\}$ and every clustering $\mathcal{C}$,*

$$\text{Top}_k(f_x) \leq \left(1 + \frac{4}{\beta} + \frac{2}{1 - \beta}\right) \text{Top}_k(y_{\mathcal{C}}).$$

Consequently,

$$\text{Top}_k(f_x) \leq \left(1 + \frac{4}{\beta} + \frac{2}{1 - \beta}\right) \text{OPT}_k^{\text{top}} \quad (k = 1, \dots, n).$$

Proof. The five pair classes partition all incidences contributing to $\sum_{u \in U} f_x(u)$. Add the direct estimate (8), Proposition 4.3, and Proposition 4.4. The second assertion follows by minimizing over the comparison clustering $\mathcal{C}$. $\square$

The coefficient

$$R(\beta) := 1 + \frac{4}{\beta} + \frac{2}{1 - \beta}$$

is minimized when

$$-\frac{4}{\beta^2} + \frac{2}{(1 - \beta)^2} = 0, \quad \text{equivalently} \quad \frac{\beta}{1 - \beta} = \sqrt{2}.$$

Thus the minimizing value is

$$\beta = 2 - \sqrt{2},$$

and

$$R(2 - \sqrt{2}) = 7 + 4\sqrt{2}. \quad (26)$$

5 Proof of the main theorem

Proof of Theorem 1.1. Choose $\beta = 2 - \sqrt{2}$, construct the metric x by (5), and apply Lemma 2.3 to obtain one clustering $\hat{\mathcal{C}}$. Coordinatewise domination implies that, for every k ,

$$\begin{aligned} \text{Top}_k(y_{\hat{\mathcal{C}}}) &\leq 5 \text{Top}_k(f_x) \\ &\leq 5(7 + 4\sqrt{2}) \text{OPT}_k^{\text{top}}. \end{aligned}$$

Now fix an arbitrary symmetric norm Φ and let $\mathcal{C}_\Phi$ be a clustering attaining OPT_Φ . Since $\text{OPT}_k^{\text{top}} \leq \text{Top}_k(y_{\mathcal{C}_\Phi})$, the preceding inequality gives

$$\text{Top}_k(y_{\hat{\mathcal{C}}}) \leq (35 + 20\sqrt{2}) \text{Top}_k(y_{\mathcal{C}_\Phi}) \quad (k = 1, \dots, n).$$

Lemma 2.2 therefore yields

$$\Phi(y_{\hat{\mathcal{C}}}) \leq (35 + 20\sqrt{2}) \Phi(y_{\mathcal{C}_\Phi}) = (35 + 20\sqrt{2}) \text{OPT}_\Phi.$$

The construction of x and the rounding algorithm are independent of Φ , so the same clustering works simultaneously for every symmetric norm. Specializing to $\Phi = \|\cdot\|_p$ proves the all- ℓ_p statement.

All steps are polynomial-time: the positive neighborhoods, retained graph, and overlap metric can be constructed by direct enumeration, and the rounding procedure has at most n rounds. $\square$

Remark 5.1 (Exact approximation factor). The final factor is

$$35 + 20\sqrt{2} \approx 63.2842712475 < 63.285.$$

Remark 5.2 (What is and is not established). The proof establishes the formal theorem in full, including an explicit polynomial-time construction; there is no unresolved mathematical step and no restriction to a subfamily of p values. Its research status is nevertheless expository rather than novel: the existence of a simultaneous all- ℓ_p approximation was already known from Davies–Moseley–Newman [2], and the stronger top- k /symmetric-norm construction used here is due to Cao–Li–Ye [1].

References

- [1] N. Cao, S. Li, and J. Ye, Simultaneously approximating all norms for massively parallel correlation clustering, in *52nd International Colloquium on Automata, Languages, and Programming (ICALP 2025)*, LIPIcs 334, Article 40, 2025. DOI: <https://doi.org/10.4230/LIPIcs.ICALP.2025.40>; preprint: <https://arxiv.org/abs/2410.09321>.
- [2] S. Davies, B. Moseley, and H. Newman, Simultaneously approximating all ℓ_p -norms in correlation clustering, in *51st International Colloquium on Automata, Languages, and Programming (ICALP 2024)*, LIPIcs 297, Article 52, 2024. DOI: <https://doi.org/10.4230/LIPIcs.ICALP.2024.52>; preprint: <https://arxiv.org/abs/2308.01534>.
- [3] S. Kalhan, K. Makarychev, and T. Zhou, Correlation clustering with local objectives, in *Advances in Neural Information Processing Systems 32*, 2019. Preprint: <https://arxiv.org/abs/1902.10829>.